Current Trends, Ethics

1. How are modern software systems changing from traditional code-centric development to
   cloud-native, API-based and continuously delivered products? Briefly describe.

Modern software systems are changing from traditional **code-centric development** to **cloud-native, API-based, and continuously delivered products**.

- **Cloud-native:** Applications are built to use cloud technologies, making them more **scalable, resilient, portable, and easier to manage**. Containers, Kubernetes, and serverless technologies are commonly used. Current Trends
- **API-based:** Modern systems use APIs so different applications and services can **communicate and share data easily**. Current Trends
- **Continuous delivery:** Using CI/CD, code is frequently integrated, automatically tested, and then delivered or deployed more quickly.

2. Compare the API protocols REST and SOAP.

REST vs SOAP

REST	SOAP
REST stands for Representational State Transfer.	**SOAP** stands for Simple Object Access Protocol.
It requires **low bandwidth**.	It requires **high bandwidth**.
It can use different response formats such as **plain text, HTML, XML, and JSON**.	It works only with **XML** format.
RESTful services follow rules such as **client-server, stateless, cacheable, layered system, and uniform interface**.	The slide does not list these REST-style constraints for SOAP.
It is generally more lightweight.	It is comparatively heavier.

3. Explain three key characteristics of responsible and ethical AI in the context of software engineering.

Three key characteristics of **Responsible and Ethical AI** in software engineering are:

1. **Fairness** – AI systems should avoid bias and make decisions fairly for different users. Current Trends
2. **Transparency** – AI decisions should be understandable and explainable to users and developers. Current Trends
3. **Accountability** – Developers should take responsibility for the outputs and impacts of AI systems.

4. Explain the difference between Native app and Web app using suitable examples.

Native App vs Web App

Native App	Web App
A **Native App** is developed for a specific operating system or device.	A **Web App** runs through a web browser.
It usually needs to be installed on the device.	It usually does not need installation.
It can use device features more directly, such as camera, GPS, and notifications.	It mainly depends on the browser and internet connection.
It is often faster and gives better device-specific performance.	It is easier to access from different devices.
Example: an iOS app built with **Swift** or an Android app built with **Kotlin**.	Example: an online shopping website opened in Chrome or Safari.

Ethics

1. Public

This means: **Protect people first.**

A software engineer should make sure the software does not harm the public, should report dangers, and should approve only safe and properly tested software.

Example:

If a banking app has a serious security bug, the engineer should not hide it.

Easy memory:

Public = People's safety first

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2. Client and Employer

This means: **Be loyal and honest to the client/employer, but not against public interest.**

Engineers should keep information confidential, use property only with permission, work only where they are competent, and warn if a project may fail.

Example:

Do not leak your company's private source code.

Easy memory:

Client & Employer = Loyalty + Confidentiality

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3. Product

This means: **Make high-quality software.**

Software should have proper requirements, realistic estimates, good documentation, testing, debugging, and review.

Example:

Before releasing an app, test it properly and fix major bugs.

Easy memory:

Product = Quality software

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4. Judgment

This means: **Make honest and independent professional decisions.**

Do not take secret payments, do not use deceptive financial practices, and disclose conflicts of interest.

Example:

Do not recommend a product just because someone secretly paid you.

Easy memory:

Judgment = Be honest and unbiased

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5. Management

This means: **Managers should manage people ethically.**

They should assign work based on skill, pay fairly, explain employment conditions honestly, and should not force employees to break ethical rules.

Example:

Do not give a very difficult security task to someone who has no security knowledge.

Easy memory:

Management = Fair and ethical leadership

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6. Profession

This means: **Protect the reputation of software engineering as a profession.**

Engineers should act responsibly, promote ethical behavior, work with reputable people, and share knowledge with other professionals. Ethics

Example:

Do not do unethical work that damages trust in software engineers.

Easy memory:

Profession = Respect the profession

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7. Colleagues

This means: **Treat coworkers fairly and support them.**

Engineers should help colleagues improve, give proper credit, review work with consent, and not damage someone’s career.

Example:

If your teammate wrote important code, do not claim it as your own.

Easy memory:

Colleagues = Respect and support coworkers

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8. Self

This means: **Keep improving yourself professionally and ethically.**

Engineers should continue learning, improve their skills, understand relevant laws and policies, and follow the ethical code. Ethics

Example:

Keep learning new technologies and security practices.

Easy memory:

Self = Keep learning and stay ethical